

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CSE) (CBCS) Degree Examinations
 May-June 2025

23CSE202 – BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Summarize distributions models in NoSQL.	8	L*2	2	1
	b) Define Big data. Why Big data? Explain its various challenges.	6	L2	1	1
	c) What is Big data analytics?	6	L1	1	1
2.	a) Explain various types of NoSQL databases with example.	8	L2	2	1
	b) What is data science? Explain different terminologies used in Big data environments.	6	L2	1	1
	c) What is sharding? Explain with example.	6	L2	2	2
Unit – II					
3.	a) What is MongoDB? List and explain different datatypes in MongoDB.	8	L1	2	1
	b) Explain the difference between the insert() and save() methods in MongoDB? Explain with query examples.	6	L2	2	2
	c) Explain CAP theorem in detail.	6	L2	2	1
4.	a) How do you work with arrays in MongoDB queries? Give an example query for creation, insertion and update of array fields.	8	L2	2	2
	b) Illustrate creation and generation of unique key in MongoDB.	6	L2	2	2
	c) What is the purpose of the count(), limit() and skip() method in MongoDB, and how do you use it?	6	L1	2	2
Unit – III					
5.	a) What is Hadoop? Explain key aspects of Hadoop.	8	L2	3	1
	b) Outline the anatomy of file read operation in HDFS with a diagram.	8	L2	3	1
	c) List out real time applications using MapReduce.	4	L2	4	2
6.	a) Explain map reduce programming with mapper and reducer with various phases.	8	L2	4	1
	b) Summarize Hadoop high level architecture with a neat sketch.	8	L2	3	1
	c) Write a note on Hadoop YARN.	4	L1	3	1
Unit – IV					
7.	a) What is HBase?. Explain its storage mechanism.	8	L2	4	1
	b) Write a note on the following:				
	i) Apache Sqoop	8	L2	4	1
	ii) Apache Solr	4	L1	4	2
	c) Why use Spark for data analysis?	4	L1	4	2
8.	a) Explain briefly Hadoop subprojects Mahout, Apache Zookeeper and Ambari.	8	L2	4	1
	b) What is in-memory computing, and why is it important in Spark? Explain PySpark, and how does it enable interactive data processing?	8	L2	4	2
	c) List out the features of HBase.	4	L1	4	1

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Unit – V

9.	a)	Explain various running modes of Pig in detail.	8	L2	5	1
	b)	Summarize HIVE datatypes and its file formats.	8	L2	5	1
	c)	Write a note on Pig philosophy.	4	L1	5	1
10.	a)	What is HIVE? Explain its architecture.	8	L2	5	1
	b)	Explain relational operators and complex datatypes in Pig.	8	L2	5	1
	c)	What is HIVE Query Language? Explain its features.	4	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
